

KYURI PARK

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EDUCATION

University of Amsterdam - Computational Science Lab, Informatics Institute	Aug 2023-Present
- PhD Candidate; expected completion August 2027. - Supervisors: Vítor V. Vasconcelos and Mike Lees.	
Visiting Scholar, Stanford University - Stanford Doerr School of Sustainability	Spring 2026
Hosted by Sara Constantino; research on climate-attitude networks, belief systems, social norms, and exploratory causal discovery.	
Complex Systems Summer School, Santa Fe Institute	Summer 2026
Selected participant; interdisciplinary training and collaborative modeling projects in complex systems.	
MSc, Methodology and Statistics, Utrecht University	Jun 2023
- Cum laude, GPA 9.04/10.00. - Supervisors: Oisín Ryan and Lourens J. Waldorp.	
BSc, Psychology (Psychological Methods), University of Amsterdam	Jun 2021
- Cum laude, GPA 9.05/10.00; Honors Programme. - Supervisors: Claudia van Borkulo and Denny Borsboom.	
BSc, Business Management, University of Houston	May 2017
- Summa cum laude, GPA 3.94/4.00; specialization in Finance Management and Accounting. - Exchange semester at Mount Royal University, Calgary, Canada.	

RESEARCH INTERESTS

Complex systems | Causal discovery | Dynamical systems | Network analysis | Computational modeling | Multiscale dynamics | Mental health | Climate beliefs, behavior, and adaptation

PEER-REVIEWED PUBLICATIONS

- Park, K., Waldorp, L. J., & Vasconcelos, V. V. (2026). A mechanistic model of symptom dynamics: Implications for statistical network analyses. *Multivariate Behavioral Research*. [in press].
- Park, K., Vasconcelos, V. V., & Lees, M. (2026). A tutorial on causal network simulation and exploration using the *causalnet* R package. *Behavior Research Methods*, 58, 204. doi:10.3758/s13428-026-03030-z.
- Park, K., Li, X., Waldorp, L. J., Lees, M., & Vasconcelos, V. V. (2026). The role of feedback loops in dynamical symptom networks. *Scientific Reports*, 16, 11273. doi:10.1038/s41598-026-38747-6.
- Park, K., Elsenburg, L. K., Nicolaou, M., Stronks, K., & Vasconcelos, V. V. (2026). Connecting precariousness and depression: From causal discovery to intervention simulation. *SSM - Mental Health*, 9, 100637. doi:10.1016/j.ssmmh.2026.100637.
- Elsenburg, L. K., Stronks, K., Galenkamp, H., Lakerveld, J., Lok, A., Park, K., Vasconcelos, V. V., & Nicolaou, M. (2026). Precariousness and depressed mood: A network analysis in the multi-ethnic HELIUS study. *Social Psychiatry and Psychiatric Epidemiology*. doi:10.1007/s00127-026-03072-w.
- Park, K., Waldorp, L. J., & Ryan, O. (2024). Discovering cyclic causal models in psychological research. *advances.in/psychology*, 2(1), e72425. doi:10.56296/aip00012.

Preprint

- Lee, S. Y., Hwang, H., Kim, T., Kim, Y., Park, K., Yoo, J., Borsboom, D., & Shin, K. (2025). Emergence of psychopathological computations in large language models. arXiv. doi:10.48550/arXiv.2504.08016.

Selected research in progress

- Slow-fast dynamics and apparent between-group differences in symptom-network structure.
- Computational modeling of executive-function engagement across development.
- Comparative climate-attitude networks and exploratory causal discovery across Japan, Europe, and the United States.
- Institutionalized category convergence in heterogeneous rate-distortion agent-based models.
- Post-disaster population displacement, migration, and urban adaptation.

GRANTS, FELLOWSHIPS, AND AWARDS

- Van der Gaag Grant, Royal Netherlands Academy of Arts and Sciences (KNAW) - PI, 2026-2027.
- Bridge Grant, Complex Systems Society, 2026.
- Energy Transition through the Lens of the SDGs (ENLENS) Research Grant, 2025-2027.
- Travel Award, International Meeting of the Psychometric Society, 2025 and 2026.
- Amsterdam University Fund - PhD Grant, 2025.
- Utrecht Excellence Scholarship, Utrecht University, 2021-2023.
- Scholarship Excellence Award, University of Houston, 2017
- UH Academic Excellence Scholarship, University of Houston, 2014-2017.

SELECTED PRESENTATIONS

- Park, K. (Jul 2026). A slow-fast perspective on between-group differences in symptom networks. International Meeting of the Psychometric Society, Seoul, South Korea.
- Park, K. (Jul 2025). Computational modeling of symptom dynamics and their implications for statistical networks. International Conference on Computational Science, Singapore.
- Park, K. (Jul 2025). Unraveling symptom dynamics: A mechanistic approach to feedback loops and causal discovery. International Meeting of the Psychometric Society, Minneapolis, MN.

INVITED TALKS AND SEMINARS

- Park, K. (May 2026). Executive function engagement as a slow-fast stochastic dynamical system. Center for Mind and Brain, University of California, Davis, Davis, CA, USA.

- Park, K. (Apr 2026). Complex systems approaches to mental health and climate-change attitudes. Biohealth Convergence and Open Sharing System (COSS), South Korea.
- Park, K. (Feb 2026). Slow-fast coupling: Curie-Weiss symptoms under context dynamics. 3rd Joint Kobe University-UvA Workshop on Computational Social Science and Intelligent Systems, Amsterdam.
- Park, K. (Jan 2026). Structural risk shifts symptom thresholds without altered symptom coupling: A slow-fast perspective. Theory Methods Lab Seminar, Amsterdam.
- Park, K. (Mar 2025). Computational approaches to mental health at the symptom level. 2nd Joint Kobe University-UvA Workshop on Computational Social Science and Intelligent Systems, Amsterdam.

RESEARCH EXPERIENCE

University of Amsterdam - Computational Science Lab, Informatics Institute	Aug 2023-Present
• Develop causal and dynamical models of complex human systems, with applications in mental health, climate attitudes, belief dynamics, and urban adaptation. • Combine causal discovery, network analysis, agent-based modeling, and simulation to study feedback processes, structural uncertainty, and intervention effects. • Lead developer of <code>causalnet</code>, an open-source R package for generating alternative directed structures from a network skeleton, including networks with cycles, summarizing their properties, and simulating nonlinear or user-defined dynamics.	
Utrecht University	Jun 2021-Aug 2023
• Enhanced the R package <code>mHMMbayes</code> (Multilevel Hidden Markov Models Using Bayesian Estimation) and developed tutorial materials. • Contributed to work on the <code>mice</code> R package and developed a synthetic-data approach for evaluating imputation-model validity with Gerko Vink. • Developed procedures for producing reproducible synthetic-data extracts from Statistics Netherlands data and co-developed the <code>synthpop.extract</code> R package with Erik-Jan van Kesteren. • Translated a Bayesian piecewise-linear regression implementation for dynamic network construction from MATLAB to R. • Applied and compared cyclic causal-discovery algorithms (CCD, FCI, CCI) for psychological data. • Coded methodological characteristics of psychological network studies in NVivo with Noémi Schuurman.	
University of Amsterdam - Psychological Methods	Nov 2019-Jan 2021
• Conducted a meta-analysis of predictors of science skepticism, developed <code>JASP</code> tutorials, and modeled networks containing variables operating at different timescales using Ising and mixed graphical models.	

TEACHING AND STUDENT MENTORING

University of Amsterdam	
• Teaching Assistant, <i>Complexity: Can It Be Simplified?</i> (BSc Honors/MSc), Vítor V. Vasconcelos, 2023–2025. • Teaching Assistant, <i>Causality</i> (MSc Artificial Intelligence), Sara Magliacane, 2024–2025. • Teaching Assistant, <i>Modeling System Dynamics</i> (MSc Information Studies), Debraj Roy, 2023–2024. • Teaching Assistant, <i>AI and Psychology</i> (BSc Psychology), Han van der Maas, 2021.	
Utrecht University	
• Developed practical materials for Python programming, Bayesian networks, relational-event models, missing data, imputation, and text mining courses; provided R programming support to sociology graduate students.	
University of Pennsylvania	
• Teaching Assistant, Social Economics: An Economic Perspective on Human Behavior, Femida Handy, Summer 2016.	
MSc Student Supervision	
• Henry Zwart, MSc in Computational Science (2026). Thesis: <i>Interventions on Asymmetric Belief Networks</i>. • Paraskevas Leivadarios, MSc in Data Science (2025). Thesis: <i>Elections and climate attitudes: how do people’s views on climate change and related policies change during an election?</i> • Gabriela Torres, MSc in Data Science (2025). Thesis: <i>The Impact of COVID-19 Interventions and Extreme Weather Harm on Support for Climate Policy</i>. • Xinhai Li, MSc in Computational Science (2024). Thesis: <i>Feedback Loops in the Dynamics of Major Depressive Disorder</i>.	

ADDITIONAL EXPERIENCE

SERVICE AND LEADERSHIP	
University of Amsterdam	2018-2021
• International Student Ambassador and member of the Psychological Methods Student Faction; represented students in programme meetings and organized academic, orientation, and outreach events.	
Utrecht University - Human Data Science Group	2021-2023
• Organized research and data-science reading groups, helped coordinate Data Donation Day, and maintained the group website.	
PROFESSIONAL EXPERIENCE	
Emirates Airline, Dubai, UAE	May 2017-Mar 2018
• Cabin crew	
Amerril Energy LLC, Houston, TX, USA	Jun 2016-Feb 2017
• Accounting assistant	
VOLUNTEER EXPERIENCE	
World Vision	2011-Present
• Sponsor since 2011; previously translated sponsor correspondence between Korean and English.	
Stray Dogs Center (SDC), Dubai, UAE	Nov 2017-Jan 2018
• Volunteered at the shelter and supported fundraising for veterinary expenses.	

METHODS, PROGRAMMING, AND LANGUAGES

Methods: Causal discovery, network analysis, dynamical-systems modeling, stochastic differential equations, agent-based modeling, simulation studies, psychometric modeling, and Bayesian and multilevel modeling.	
Programming: R, Python, MATLAB; JavaScript (learning), C++ (learning)	
Languages: Korean (native), English (fluent), Dutch (B1).	